Chapter 4. Nonstationary Time Series Models and Model Selection

ARIMA, unit root tests, model selection and diagnostics

Contents

4.1 Autoregressive integrated moving average models **2**

4.2 Unit root tests **2**

4.2.1 The AR(1) case 3

4.2.2 The AR(p) case: augmented Dickey–Fuller test 5

4.3 Model selection and diagnostics **5**

4.3.1 The Box–Jenkins method 5

4.3.2 Model selection criteria 6

4.3.3 Model diagnostics 6

4.4 Selected exercises **7**

4.1 Autoregressive integrated moving average models

Motivation. Series that wander without a fixed level are differenced until stationary and then fitted by ARMA: the ARIMA class.

The *random walk*

$$X_t = X_{t-1} + \epsilon_t, \qquad \text{equivalently} \qquad X_t = X_0 + \epsilon_1 + \cdots + \epsilon_t,$$

reads $(1-B)X_t = \epsilon_t$, i.e. $\nabla X_t = \epsilon_t$ with $\nabla = 1 - B$.

Example (the random walk is not stationary)

With $X_0 = 0$ and $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$, $EX_t = 0$ but $\text{Var}(X_t) = \text{Cov}(X_t, X_{t+h}) = t\sigma^2$ $(h \geq 0)$ depends on t, while $\nabla X_t = \epsilon_t$ is white noise; $\phi(z) = 1 - z$ has its root on the unit circle.

If $Z_t = (1-B)^d X_t$ is stationary ARMA, $\phi(B)Z_t = \theta(B)\epsilon_t$, then $\phi(B)(1-B)^d X_t = \theta(B)\epsilon_t$.

Definition 4.1.1 (ARIMA(p,d,q) model)

A time series $\{X_t\}$ follows an *autoregressive integrated moving average model* of orders p, d, q, written $\{X_t\} \sim \text{ARIMA}(p,d,q)$, if

$$\phi(B)(1-B)^d X_t = \theta(B)\epsilon_t, \qquad \{\epsilon_t\} \sim \text{IID}(0, \sigma^2),$$

where $\phi(z) = 1 - \phi_1 z - \cdots - \phi_p z^p$ and $\theta(z) = 1 + \theta_1 z + \cdots + \theta_q z^q$. In practice $d \in \{0, 1, 2\}$.

ϕ, θ are assumed causal and invertible (no zeros in $|z| \leq 1$); IID errors are needed for the inference below.

Remark 4.1.2

The random walk is ARIMA$(0,1,0)$. If the differenced series has a nonzero mean μ, the model is written

$$\phi(B)((1-B)^d X_t - \mu) = \theta(B)\epsilon_t, \qquad \mu = E(1-B)^d X_t.$$

Polynomial trends. For $X_t = t + \epsilon_t$, $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$,

$$(1-B)X_t = t + \epsilon_t - (t-1) - \epsilon_{t-1} = 1 + \epsilon_t - \epsilon_{t-1},$$

so $\{X_t\}$ is ARIMA$(0,1,1)$ with $\mu = 1$ and non-invertible $\theta(B) = 1 - B$. For $X_t = a_0 + a_1 t + \cdots + a_r t^r + \epsilon_t$, $(1-B)$ lowers the degree by one ($a\{t^k - (t-1)^k\} = ak\, t^{k-1} + \cdots$), so $(1-B)^r X_t = a_r r! + (1-B)^r \epsilon_t$ is stationary. The converse fails; a known polynomial trend is better fitted directly.

4.2 Unit root tests

Motivation. Testing for a root at $z = 1$ needs Dickey–Fuller critical values, since under a unit root LS converges at rate n to a Brownian-motion functional.

4.2.1 The AR(1) case

Let $X_t = \phi X_{t-1} + \epsilon_t$, $X_0 = 0$, $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$, and test

$$H_0 : \phi = 1 \qquad \text{vs} \qquad H_1 : |\phi| < 1.$$

With $\hat{\phi}_{LS} = \sum_{t=1}^{n} X_{t-1}X_t / \sum_{t=1}^{n} X_{t-1}^2$, let

$$T_n = n(\hat{\phi}_{LS} - 1). \tag{4.1}$$

Under H_0, $\hat{\phi}_{LS} - 1 = \sum X_{t-1}\epsilon_t / \sum X_{t-1}^2$, so

$$T_n = \frac{n^{-1}\sum_{t=1}^{n} X_{t-1}\epsilon_t}{n^{-2}\sum_{t=1}^{n} X_{t-1}^2}. \tag{4.2}$$

Define the partial-sum process

$$W_n(s) = \frac{1}{\sqrt{n}\,\sigma} \sum_{j=1}^{[ns]} \epsilon_j, \qquad 0 \le s \le 1; \tag{4.3}$$

under H_0, $X_t = \sigma\sqrt{n}\, W_n(t/n)$.

Definition 4.2.1 (Brownian motion)

A stochastic process $\{W(s) : s \ge 0\}$ with continuous sample paths is a (standard) *Brownian motion* if
(1) $W(0) = 0$;
(2) for all $t > s \ge 0$, $W(t) - W(s) \sim N(0, t-s)$;
(3) for all $0 \le t_1 < \cdots < t_k$, the increments $W(t_1)$, $W(t_2) - W(t_1), \ldots, W(t_k) - W(t_{k-1})$ are independent.

$W_n \xrightarrow{d} W$ means weak convergence in $D[0,1]$ (Skorokhod topology): $Ef(W_n) \to Ef(W)$ for bounded continuous f; then $h(W_n) \xrightarrow{d} h(W)$ for h continuous at a.e. path of W.

Theorem 4.2.2 (Functional central limit theorem)

Let $\{\epsilon_t\} \sim \text{IID}(0, \sigma^2)$ and define W_n by (4.3). Then, as $n \to \infty$, $W_n \xrightarrow{d} W$, a standard Brownian motion on $[0,1]$.

Proof. [Supplement] Let $S_k = \epsilon_1 + \cdots + \epsilon_k$, $W_n(s) = S_{[ns]}/(\sigma\sqrt{n})$. We verify finite-dimensional convergence and tightness (Prokhorov).

Step 1 (finite-dimensional laws). For $0 = s_0 < \cdots < s_k \le 1$, $m_j = [ns_j] - [ns_{j-1}]$, $m_j/n \to s_j - s_{j-1}$,

$$W_n(s_j) - W_n(s_{j-1}) = \sqrt{\frac{m_j}{n}} \cdot \frac{1}{\sigma\sqrt{m_j}} \sum_{i=[ns_{j-1}]+1}^{[ns_j]} \epsilon_i \xrightarrow{d} N(0, s_j - s_{j-1})$$

(CLT, Slutsky). The increments use disjoint blocks, hence are independent and converge jointly to independent $N(0, s_j - s_{j-1})$; summing gives $(W_n(s_1), \ldots, W_n(s_k)) \xrightarrow{d} (W(s_1), \ldots, W(s_k))$.

Step 2 (a maximal inequality). By Billingsley (Thm. 8.4) it suffices that for every $\varepsilon > 0$ there are $\lambda > 1$, n_0 with

$$P\Big(\max_{k \le n} |S_k| \ge \lambda\sigma\sqrt{n}\Big) \le \frac{\varepsilon}{\lambda^2} \qquad (n \ge n_0).$$

Etemadi: $P(\max_{k\le n}|S_k|\ge 3a)\le 3\max_{k\le n}P(|S_k|\ge a)$; take $a=\mu\sigma\sqrt{n}$, $\mu=\lambda/3$. By the CLT, $P(|S_k|\ge\mu\sigma\sqrt{k})\le 2P(|N|\ge\mu)\le 6/\mu^4$ for $k\ge k_\mu$ ($N\sim N(0,1)$, $EN^4=3$). Hence

(i) for $k_\mu\le k\le n$: $P(|S_k|\ge\mu\sigma\sqrt{n})\le P(|S_k|\ge\mu\sigma\sqrt{k})\le 6/\mu^4$;

(ii) for $k<k_\mu$: by Chebyshev, $P(|S_k|\ge\mu\sigma\sqrt{n})\le k/(\mu^2 n)\le k_\mu/(\mu^2 n)\le 1/\mu^4$ once $n\ge\mu^2 k_\mu$.

So $P(\max_{k\le n}|S_k|\ge\lambda\sigma\sqrt{n})\le 3\cdot 7/\mu^4=1701/\lambda^4\le\varepsilon/\lambda^2$ for $\lambda^2\ge 1701/\varepsilon$.

Step 3 (conclusion). Every subsequence has a weakly convergent further subsequence whose limit has the finite-dimensional laws of W; these determine the law, so $W_n\xrightarrow{d}W$. ■

Proposition (Limit of the unit-root statistics)

Under $H_0:\phi=1$, with T_n as in (4.1),

$$T_n\xrightarrow{d}T:=\frac{\frac12(W(1)^2-1)}{\int_0^1 W(s)^2\,ds}=\frac{\int_0^1 W(s)\,dW(s)}{\int_0^1 W(s)^2\,ds}.$$

Under $H_1:|\phi|<1$, $P(|T_n|\ge c)\to 1$ for every constant c; that is, the test rejecting for large $|T_n|$ is consistent.

Proof. [Supplement]

Step 1 (numerator). $X_t^2-X_{t-1}^2=2X_{t-1}\epsilon_t+\epsilon_t^2$; summing with $X_0=0$,

$$\frac1n\sum_{t=1}^n X_{t-1}\epsilon_t=\frac12\Big(\frac{X_n^2}{n}-\frac1n\sum_{t=1}^n\epsilon_t^2\Big)=\frac{\sigma^2}{2}\Big(W_n(1)^2-\frac{1}{n\sigma^2}\sum_{t=1}^n\epsilon_t^2\Big).$$

Step 2 (denominator). $W_n(s)=X_{t-1}/(\sigma\sqrt{n})$ on $[(t-1)/n,t/n)$, so

$$\frac{1}{n^2}\sum_{t=1}^n X_{t-1}^2=\sigma^2\sum_{t=1}^n\int_{(t-1)/n}^{t/n}W_n(s)^2\,ds=\sigma^2\int_0^1 W_n(s)^2\,ds.$$

Step 3 (joint limit). $w\mapsto(w(1),\int_0^1 w^2)$ is continuous at continuous w, so $(W_n(1),\int_0^1 W_n^2)\xrightarrow{d}(W(1),\int_0^1 W^2)$; with $n^{-1}\sum\epsilon_t^2\xrightarrow{p}\sigma^2$ and Slutsky, all three converge jointly.

Step 4 (ratio). As $\int_0^1 W^2>0$ a.s., $(a,b,c)\mapsto\frac12(a^2-c)/b$ is continuous at the limit, giving $T_n\xrightarrow{d}\frac12(W(1)^2-1)/\int_0^1 W^2$; Itô's formula $W(1)^2=2\int_0^1 W\,dW+1$ gives the second form.

Step 5 (consistency). If $|\phi|<1$, $T_n=n(\hat\phi_{LS}-\phi)+n(\phi-1)$ with $n(\hat\phi_{LS}-\phi)=O_p(\sqrt{n})$ (Chapter 3), so $T_n/n\xrightarrow{p}\phi-1<0$ and $P(|T_n|\ge c)\to 1$. ■

Thus the rate is n (super-consistency) and T is left-skewed, not normal. Reject when $|T_n|>c_\alpha$, $P(|T|>c_\alpha)=\alpha$, with quantiles tabulated by simulation.

Dickey–Fuller test. The studentized version is

$$DF_n=\Big(\frac{\sum_{t=1}^n X_{t-1}^2}{\hat\sigma^2}\Big)^{1/2}(\hat\phi_{LS}-1),\qquad \hat\sigma^2=\frac1n\sum_{t=1}^n(X_t-\hat\phi_{LS}X_{t-1})^2.$$

Under H_0,

$$DF_n=\frac{n^{-1}\sum_{t=1}^n X_{t-1}\epsilon_t}{\hat\sigma\,(n^{-2}\sum_{t=1}^n X_{t-1}^2)^{1/2}}\xrightarrow{d}DF:=\frac{\frac12(W(1)^2-1)}{(\int_0^1 W(s)^2ds)^{1/2}},$$

by Steps 1–3 and $\hat\sigma^2\xrightarrow{p}\sigma^2$: with $X_t-\hat\phi_{LS}X_{t-1}=\epsilon_t-(\hat\phi_{LS}-1)X_{t-1}$,

$$\hat\sigma^2=\frac1n\sum\epsilon_t^2-2(\hat\phi_{LS}-1)\cdot\frac1n\sum X_{t-1}\epsilon_t+(\hat\phi_{LS}-1)^2\cdot\frac1n\sum X_{t-1}^2,$$

and $\hat{\phi}_{LS} - 1 = O_p(n^{-1})$, $n^{-1}\sum X_{t-1}\epsilon_t = O_p(1)$, $n^{-1}\sum X_{t-1}^2 = O_p(n)$ make the last two terms $O_p(n^{-1})$. Reject when $|DF_n| \geq DF_\alpha$, $P(|DF| \geq DF_\alpha) = \alpha$.

4.2.2 The AR(p) case: augmented Dickey–Fuller test

For $\phi(B)X_t = \epsilon_t$, $\phi(B) = 1 - \phi_1 B - \cdots - \phi_p B^p$, test $\phi(1) = 0$.

Lemma (Error-correction form)

For any $\phi_1, \ldots, \phi_p$, with $\Delta X_t = X_t - X_{t-1}$,

$$X_t - \sum_{i=1}^{p} \phi_i X_{t-i} = X_t - \beta X_{t-1} - \sum_{j=1}^{p-1} \psi_j \Delta X_{t-j}, \qquad \beta = \sum_{i=1}^{p} \phi_i = 1 - \phi(1), \quad \psi_j = -\sum_{i=j+1}^{p} \phi_i.$$

In particular $\phi(1) = 0$ iff $\beta = 1$, and then $\phi(B) = \psi(B)(1 - B)$ with $\psi(B) = 1 - \psi_1 B - \cdots - \psi_{p-1} B^{p-1}$.

Proof. [Supplement] Using $X_{t-i} = X_{t-1} - \sum_{j=1}^{i-1} \Delta X_{t-j}$ and exchanging sums,

$$\sum_{i=1}^{p} \phi_i X_{t-i} = \Big(\sum_{i=1}^{p} \phi_i\Big) X_{t-1} - \sum_{i=1}^{p} \phi_i \sum_{j=1}^{i-1} \Delta X_{t-j} = \beta X_{t-1} - \sum_{j=1}^{p-1} \Big(\sum_{i=j+1}^{p} \phi_i\Big) \Delta X_{t-j}.$$

If $\beta = 1$ the right side is $\Delta X_t - \sum_j \psi_j \Delta X_{t-j} = \psi(B)(1 - B)X_t$. ■

So the AR(p) model is the regression

$$X_t = \beta X_{t-1} + \psi_1 \Delta X_{t-1} + \cdots + \psi_{p-1} \Delta X_{t-p+1} + \eta_t, \qquad \{\eta_t\} \sim \mathrm{IID}(0, \tau^2), \tag{4.4}$$

with $\beta = 1$, $\eta_t = \epsilon_t$ under the unit root. The *augmented Dickey–Fuller* (ADF) test rejects for large $|(\hat{\beta} - 1)/\,\mathrm{sd}(\hat{\beta})|$; its limit law is the same DF.

4.3 Model selection and diagnostics

Motivation. Orders are identified by sample ACF/PACF or information criteria, and the fit is checked by overfitting and residuals.

4.3.1 The Box–Jenkins method

Three stages:

(1) *identification* of p, q, (2) *estimation* (Chapter 3), (3) *diagnostic checking* (Section 4.3.3), returning to (1) on failure (Figure 1).

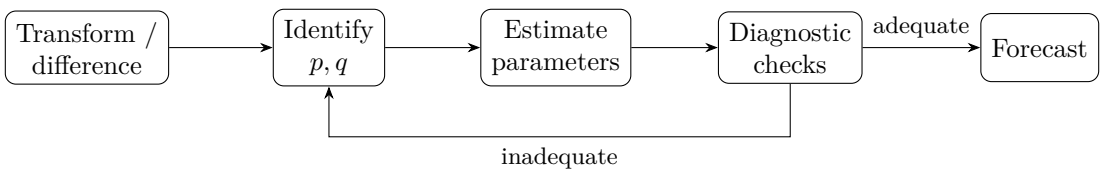


Figure 1: The Box–Jenkins modelling cycle.

Making the series stationary. Non-constant spread: Box–Cox $(y^\lambda - 1)/\lambda$ $(\lambda \neq 0)$, $\sqrt{y}$, $1/y$ or $\log y$ (standard for economic data). Trend: difference. A slowly decaying SACF indicates nonstationarity; a unit-root test (Section 4.2) decides. Usually $d \in \{0, 1, 2\}$.

Identification by ACF and PACF. Compare SACF and SPACF with Table 1: the SACF guides q, the SPACF p. By parsimony ($p + q + 2$ parameters) usually $p, q \leq 3$.

Model	ACF	PACF
AR(p)	decays exponentially or as a damped sine	cuts off after lag p
MA(q)	cuts off after lag q	decays exponentially or as a damped sine
ARMA(p, q)	decays after lag $q - p$	decays after lag $p - q$

Table 1: ACF and PACF of ARMA models (Table 4.3.1).

AR(p): for $k \geq p + 1$, $\hat{\phi}_{kk} \approx N(0, 1/n)$, so $H_0 : \phi_{kk} = 0$ is rejected when $|\hat{\phi}_{kk}| > z_{\alpha/2}/\sqrt{n}$.

MA(q): reject $H_0 : \rho(k) = 0$ when $|\hat{\rho}(k)| > z_{\alpha/2}\,\text{S.E.}(\hat{\rho}(k))$, where by Bartlett

$$\text{S.E.}(\hat{\rho}(k)) \approx \Big(\frac{1 + 2\sum_{j=1}^{q} \hat{\rho}(j)^2}{n}\Big)^{1/2}, \qquad k \geq q + 1;$$

if no $k \geq q+1$ rejects, MA(q) is tentatively chosen. Mixed ARMA orders are chosen by the criteria below.

4.3.2 Model selection criteria

Consider the nested models

$$M_1 = \{N(\mu, 1) : -\infty < \mu < \infty\}, \qquad M_2 = \{N(\mu, \sigma^2) : -\infty < \mu < \infty,\ 0 < \sigma^2 < \infty\}.$$

Choosing M_2 when M_1 holds wastes a parameter; the reverse omits an essential one, which is worse. With maximized likelihoods $MLE_1 = f_1(X_1, \ldots, X_n; \hat{\mu})$, $MLE_2 = f_2(X_1, \ldots, X_n; \hat{\mu}, \hat{\sigma}^2)$, select the smaller of

$$AIC(M_1) = -2\log(MLE_1) + 2 \times 1, \qquad AIC(M_2) = -2\log(MLE_2) + 2 \times 2.$$

In general

$$AIC(\text{model}) = -2\log(MLE) + 2\,(\text{number of parameters in the model}), \tag{4.5}$$

$$BIC(\text{model}) = -2\log(MLE) + \log(n)\,(\text{number of parameters in the model}).$$

The likelihood always grows with the number of parameters; the penalty enforces parsimony. AIC estimates the expected Kullback–Leibler discrepancy and applies directly to ARMA models.

[Supplement] AIC versus BIC

For a true model M and $M' \supset M$ with one extra parameter, $\Lambda_n = 2\log(MLE') - 2\log(MLE) \xrightarrow{d} \chi^2_1$. AIC picks M' iff $\Lambda_n > 2$ (limit probability $P(\chi^2_1 > 2) \approx 0.157$); BIC iff $\Lambda_n > \log n$ (probability $\to 0$). If M is false and M' true, $\Lambda_n/n \to$ a positive KL divergence, so both reject M. Hence BIC is consistent, AIC overfits but is efficient for prediction.

4.3.3 Model diagnostics

An adequate model is used for forecasting; otherwise return to identification.

(1) Overfitting. For a tentative ARIMA(p,d,q), fit ARIMA$(p+1,d,q)$ and ARIMA$(p,d,q+1)$ and check:

(a) are the added parameters judged to be zero?

(b) are the estimates of the original parameters essentially unchanged?

E.g. for MA(1), fit ARMA(1,1) and MA(2): ϕ_1, θ_2 should be insignificant (asymptotically normal with mean 0 under the tentative model) and the three $\hat{\theta}_1$ should agree.

(2) Residual analysis. For the stationary ARMA(p,q) model

$$\phi(B)(X_t - \mu) = \theta(B)\epsilon_t, \tag{4.6}$$

the residuals

$$e_t = \hat{\theta}(B)^{-1}\hat{\phi}(B)(X_t - \hat{\mu}), \qquad t = 1, \ldots, n,$$

(with $X_t = 0$ for $t \le 0$) should behave like IID errors. Check:

(a) zero mean ($\bar{e}$ vs its standard error), (b) constant variance (residual plot), (c) independence, via

$$\rho_e(k) = \frac{\sum_{t=1}^{n-k}(e_t - \bar{e})(e_{t+k} - \bar{e})}{\sum_{t=1}^{n}(e_t - \bar{e})^2}.$$

Under H_0: "(4.6) is true", $\mathrm{Var}(\rho_e(k)) \approx 1/n$; reject when $|\rho_e(k)| \ge z_{\alpha/2}\,\mathrm{S.E.}(\rho_e(k))$. An ARMA fit to the residuals should give ARMA(0,0).

Lag-by-lag tests inflate the overall type I error; the simultaneous *portmanteau* (*Ljung–Box*) statistic

$$Q = n(n+2)\sum_{h=1}^{K}\frac{\rho_e^2(h)}{n-h}$$

is approximately $\chi^2_{K-p-q-1}$ under H_0; reject when Q exceeds its upper $100\alpha\%$ point.

The *Durbin–Watson* statistic tests lag-one correlation against AR(1) errors:

$$DW = \frac{\sum_{t=2}^{n}(e_t - e_{t-1})^2}{\sum_{t=1}^{n} e_t^2} \approx 2(1 - \rho_e(1)),$$

as the numerator is $\sum_{t\ge2} e_t^2 + \sum_{t\ge2} e_{t-1}^2 - 2\sum_{t\ge2} e_t e_{t-1} \approx 2\sum e_t^2 - 2\sum e_t e_{t-1}$. So $0 \le DW \le 4$; $DW \approx 2$: none, < 2: positive, > 2: negative correlation (tables: Durbin and Watson, 1971).

4.4 Selected exercises

Motivation. Exercises on differencing trends, the random walk, Brownian motion and over-differencing.

Exercise 4.1 (Differencing a polynomial trend)

Let $X_t = a_0 + a_1 t + \cdots + a_r t^r + \epsilon_t$ with $r \ge 1$, $a_r \ne 0$, $\{\epsilon_t\} \sim WN(0, \sigma^2)$. Show that
(a) $(1-B)^r t^r = r!$;
(b) $(1-B)^r \epsilon_t = \sum_{j=0}^{r}\binom{r}{j}(-1)^j \epsilon_{t-j}$;
(c) $(1-B)^r X_t$ is stationary, and find its mean and variance.

Solution.

(a)

Step 1. For degree $k \geq 1$, by the binomial theorem

$$(1-B)t^k = t^k - (t-1)^k = -\sum_{i=0}^{k-1} \binom{k}{i}(-1)^{k-i}t^i = k\,t^{k-1} + (\text{terms of degree} \leq k-2),$$

so $(1-B)$ maps degree k, leading coefficient c, to degree $k-1$, leading coefficient kc; $(1-B)c = 0$.

Step 2. By induction, $(1-B)^j t^r$ has degree $r-j$ and leading coefficient $r(r-1)\cdots(r-j+1)$; $j = r$ gives $r!$.

(b) 1 and B commute, so $(1-B)^r = \sum_{j=0}^{r}\binom{r}{j}(-1)^j B^j$; apply to ϵ_t.

(c)

Step 1. For $k < r$, $(1-B)^r t^k = (1-B)^{r-k}k! = 0$. By (a),(b),

$$Z_t := (1-B)^r X_t = a_r\, r! + \sum_{j=0}^{r} c_j \epsilon_{t-j}, \qquad c_j = \binom{r}{j}(-1)^j.$$

Step 2. As in (2.15), $\gamma_Z(h) = \sigma^2 \sum_{j=0}^{r-|h|} c_j c_{j+|h|}$ for $|h| \leq r$, 0 otherwise: $\{Z_t\}$ is stationary with $EZ_t = a_r\, r!$.

Step 3. By Vandermonde,

$$\mathrm{Var}(Z_t) = \sigma^2 \sum_{j=0}^{r}\binom{r}{j}^2 = \binom{2r}{r}\sigma^2.$$

■

Exercise 4.2 (Markov property of the random walk)

Let $X_t = X_{t-1} + \epsilon_t$, $X_0 = 0$, $\{\epsilon_t\} \sim IID(0, \sigma^2)$ with values in a countable set. Show that, whenever the conditioning events have positive probability,

$$P(X_n = i_n \mid X_0 = i_0, X_1 = i_1, \ldots, X_{n-1} = i_{n-1}) = P(X_n = i_n \mid X_{n-1} = i_{n-1}).$$

Solution.

Step 1. $X_0, \ldots, X_{n-1}$ are functions of $\epsilon_1, \ldots, \epsilon_{n-1}$, hence independent of ϵ_n; so are $A = \{X_0 = i_0, \ldots, X_{n-1} = i_{n-1}\}$ and $A' = \{X_{n-1} = i_{n-1}\}$.

Step 2. On A, $X_n = i_{n-1} + \epsilon_n$, so

$$P(X_n = i_n, A) = P(\epsilon_n = i_n - i_{n-1}, A) = P(\epsilon_n = i_n - i_{n-1})P(A),$$

i.e. $P(X_n = i_n \mid A) = P(\epsilon_1 = i_n - i_{n-1})$.

Step 3. Likewise $P(X_n = i_n \mid A') = P(\epsilon_1 = i_n - i_{n-1})$.

For general ϵ_t with d.f. F, the freezing lemma gives $P(X_n \leq x \mid X_0, \ldots, X_{n-1}) = F(x - X_{n-1}) = P(X_n \leq x \mid X_{n-1})$. ■

Exercise 4.3 (Quadratic trend plus a local linear trend)

Let

$$X_t = \alpha_0 + \alpha_1 t + \alpha_2 t^2 + \mu_t + \epsilon_t, \qquad \mu_t = \mu_{t-1} + \beta_t + \eta_t, \qquad \beta_t = \beta_{t-1} + \zeta_t,$$

where $\{\epsilon_t\} \sim WN(0, \sigma_\epsilon^2)$, $\{\eta_t\} \sim WN(0, \sigma_\eta^2)$, $\{\zeta_t\} \sim WN(0, \sigma_\zeta^2)$ are mutually independent. Difference $\{X_t\}$ to obtain a stationary series and find its mean and variance.

Solution.

Step 1. $(1-B)\beta_t = \zeta_t$ and $(1-B)\mu_t = \beta_t + \eta_t$, so

$$(1-B)^2\mu_t = (1-B)\beta_t + (1-B)\eta_t = \zeta_t + \eta_t - \eta_{t-1}.$$

Step 2. By Exercise 4.1, $(1-B)^2(\alpha_0 + \alpha_1 t + \alpha_2 t^2) = 2\alpha_2$, hence

$$D_t := (1-B)^2 X_t = 2\alpha_2 + \zeta_t + (\eta_t - \eta_{t-1}) + (\epsilon_t - 2\epsilon_{t-1} + \epsilon_{t-2}).$$

Step 3. The three uncorrelated finite MAs have ACVFs (2.15) (coefficients (1), $(1,-1)$, $(1,-2,1)$); by Exercise 1.7, $\{D_t\}$ is stationary with

$$\gamma_D(0) = \sigma_\zeta^2 + 2\sigma_\eta^2 + 6\sigma_\epsilon^2, \quad \gamma_D(\pm 1) = -\sigma_\eta^2 - 4\sigma_\epsilon^2, \quad \gamma_D(\pm 2) = \sigma_\epsilon^2,$$

and $\gamma_D(h) = 0$ for $|h| \geq 3$.

Step 4. Hence $ED_t = 2\alpha_2$ and $\mathrm{Var}(D_t) = \sigma_\zeta^2 + 2\sigma_\eta^2 + 6\sigma_\epsilon^2$.

One difference is not enough: $(1-B)X_t = \alpha_1 - \alpha_2 + 2\alpha_2 t + \beta_t + \eta_t + \epsilon_t - \epsilon_{t-1}$ has a linear mean and contains the random walk β_t. ■

Exercise 4.8 (Properties of Brownian motion)

Let $\{W(s) : s \geq 0\}$ be a standard Brownian motion (Definition 4.2.1). Show that
(a) $EW(t)W(s) = t \wedge s$;
(b) $Z(s) = W(\eta s)/\sqrt{\eta}$ is a Brownian motion for every $\eta > 0$;
(c) $G(s) = sW(1/s)$ for $s > 0$, $G(0) = 0$, is a Brownian motion.

Solution.

(a) For $s \leq t$, $W(s) \perp W(t) - W(s)$, both mean 0, $EW(s)^2 = s$: $EW(t)W(s) = EW(s)^2 + E[W(s)(W(t) - W(s))] = s$.

Characterization. A process with continuous paths, $G(0) = 0$, and jointly normal mean-zero finite-dimensional laws with $\mathrm{Cov}(G(s), G(t)) = s \wedge t$ is a Brownian motion; conversely W has these properties.

Step 1. For $t > s$, $G(t) - G(s) \sim N(0,\, t - 2(s \wedge t) + s) = N(0, t-s)$.

Step 2. For $t_0 < \cdots < t_k$ the increments Δ_i are jointly normal and, for $i < j$ $(t_i \leq t_{j-1})$,

$$\mathrm{Cov}(\Delta_i, \Delta_j) = t_i \wedge t_j - t_i \wedge t_{j-1} - t_{i-1} \wedge t_j + t_{i-1} \wedge t_{j-1} = t_i - t_i - t_{i-1} + t_{i-1} = 0,$$

hence independent (Exercise 1.6); $t_0 = 0$ gives (3).

Step 3. Conversely, $(W(t_1), \ldots, W(t_k))$ is a linear image of independent normal increments, with covariance $s \wedge t$ by (a).

(b) Z is continuous, $Z(0) = 0$, jointly normal, and $\mathrm{Cov}(Z(s), Z(t)) = (\eta s \wedge \eta t)/\eta = s \wedge t$.

(c)

Step 1. $\mathrm{Cov}(G(s), G(t)) = st\,(1/s \wedge 1/t) = s \wedge t$; laws are jointly normal; paths are continuous on $(0, \infty)$.

Step 2. Continuity at 0 means $W(u)/u \to 0$ a.s.; the SLLN gives $W(n)/n \to 0$.

Step 3. $M_k = \sup_{u \in [k, k+1]} |W(u) - W(k)|$ has the law of $M_0 = \sup_{u \leq 1} |W(u)|$ (same Gaussian covariance), and Doob's L^2 inequality gives $EM_0^2 \leq 4EW(1)^2 = 4$, so $EM_0 \leq 2$.

Step 4. $\sum_{k \geq 1} P(M_k > \varepsilon k) = \sum_{k \geq 1} P(M_0/\varepsilon > k) \leq E(M_0)/\varepsilon < \infty$; Borel–Cantelli gives $M_k/k \to 0$ a.s.

Step 5. For $u \in [k, k+1]$, $|W(u)|/u \le (|W(k)| + M_k)/k \to 0$, so $G(s) \to 0$ as $s \downarrow 0$; the characterization applies. ■

Exercise 4.9 (Over-differencing an MA(2))

Let $X_t = \epsilon_t + \theta_1\epsilon_{t-1} + \theta_2\epsilon_{t-2}$, $\{\epsilon_t\} \sim IID\, N(0, \sigma^2)$, be stationary, and suppose one fits $(1-B)X_t = Y_t$. Find the ACVF of $\{Y_t\}$.

Solution.

Step 1. $Y_t = \epsilon_t + (\theta_1 - 1)\epsilon_{t-1} + (\theta_2 - \theta_1)\epsilon_{t-2} - \theta_2\epsilon_{t-3}$, an MA(3) with $c_0 = 1$, $c_1 = \theta_1 - 1$, $c_2 = \theta_2 - \theta_1$, $c_3 = -\theta_2$.

Step 2. By Exercise 2.9(a) with $c = 1$, $\gamma_Y(h) = 2\gamma_X(h) - \gamma_X(h-1) - \gamma_X(h+1)$, with $\gamma_X(0) = (1 + \theta_1^2 + \theta_2^2)\sigma^2$, $\gamma_X(\pm 1) = (\theta_1 + \theta_1\theta_2)\sigma^2$, $\gamma_X(\pm 2) = \theta_2\sigma^2$:

$$\begin{aligned} \gamma_Y(0) &= 2\sigma^2(1 + \theta_1^2 + \theta_2^2 - \theta_1 - \theta_1\theta_2), \\ \gamma_Y(\pm 1) &= \sigma^2(2\theta_1 + 2\theta_1\theta_2 - \theta_2 - 1 - \theta_1^2 - \theta_2^2), \\ \gamma_Y(\pm 2) &= \sigma^2(2\theta_2 - \theta_1 - \theta_1\theta_2), \qquad \gamma_Y(\pm 3) = -\theta_2\sigma^2, \end{aligned}$$

and $\gamma_Y(h) = 0$ for $|h| \ge 4$ (agreeing with $\sigma^2 \sum_j c_j c_{j+|h|}$).

Step 3. The MA polynomial $(1 - z)\theta(z)$ vanishes at $z = 1$: over-differencing destroys invertibility. For $\theta_1 = \theta_2 = 0$, $\rho_Y(1) = -\frac{1}{2}$; a lag-one SACF near -0.5 warns of unnecessary differencing. ■